

ECON 108

INTRODUCTION TO ECONOMICS

INSTRUCTOR: ALEJANDRO MELO PONCE

The Firm: Cost and Output Determination

CHAPTER XI (from the book)

Chapter Outline

Short Run versus Long Run

A Firm's Production

Short-Run Costs to the Firm

Long-Run Cost Curves

Short Run versus Long Run

Short run

- A time period when at least one input, such as plant size, cannot be changed
- **Plant size**
 - The physical size of the factories that a firm owns and operates to produce its output

Short Run versus Long Run (cont'd)

Long run

- The time period in which all factors of production can be varied

Short Run versus Long Run (cont'd)

Managers take account of both the short-run and long-run consequences of their behavior.

While making decisions about what to do today, tomorrow, and next week, they keep an eye on the long-run benefits.

A Firm's Production

A firm takes numerous inputs, combines them using a technological production process, and ends up with output.

We classify production inputs in two broad categories: labor and capital.

A Firm's Production (cont'd)

Output per time period = Some function of capital and labor inputs

Or:

$$Q = f(K, L)$$

Q = output / time period

K = capital

L = labor

A Firm's Production (cont'd)

Production

- Any activity that results in the conversion of resources into products that can be used in consumption

A Firm's Production (cont'd)

Production function

- The production function is the relationship between maximum physical output and the quantity of capital and labor used in the production process.
- The production function is a technological relationship between inputs and output.

Example: Using a Virus to Produce Lithium Batteries with Fewer Inputs

Battery manufacturers have developed a lithium-air design, which enables an electric current to be generated by combining oxygen with lithium.

This design reduces the amount of lithium required to produce each battery.

As a result, more batteries can be produced with a given amount of inputs.

A Firm's Production (cont'd)

Average physical product

- Total product divided by the variable input

A Firm's Production (cont'd)

Marginal physical product

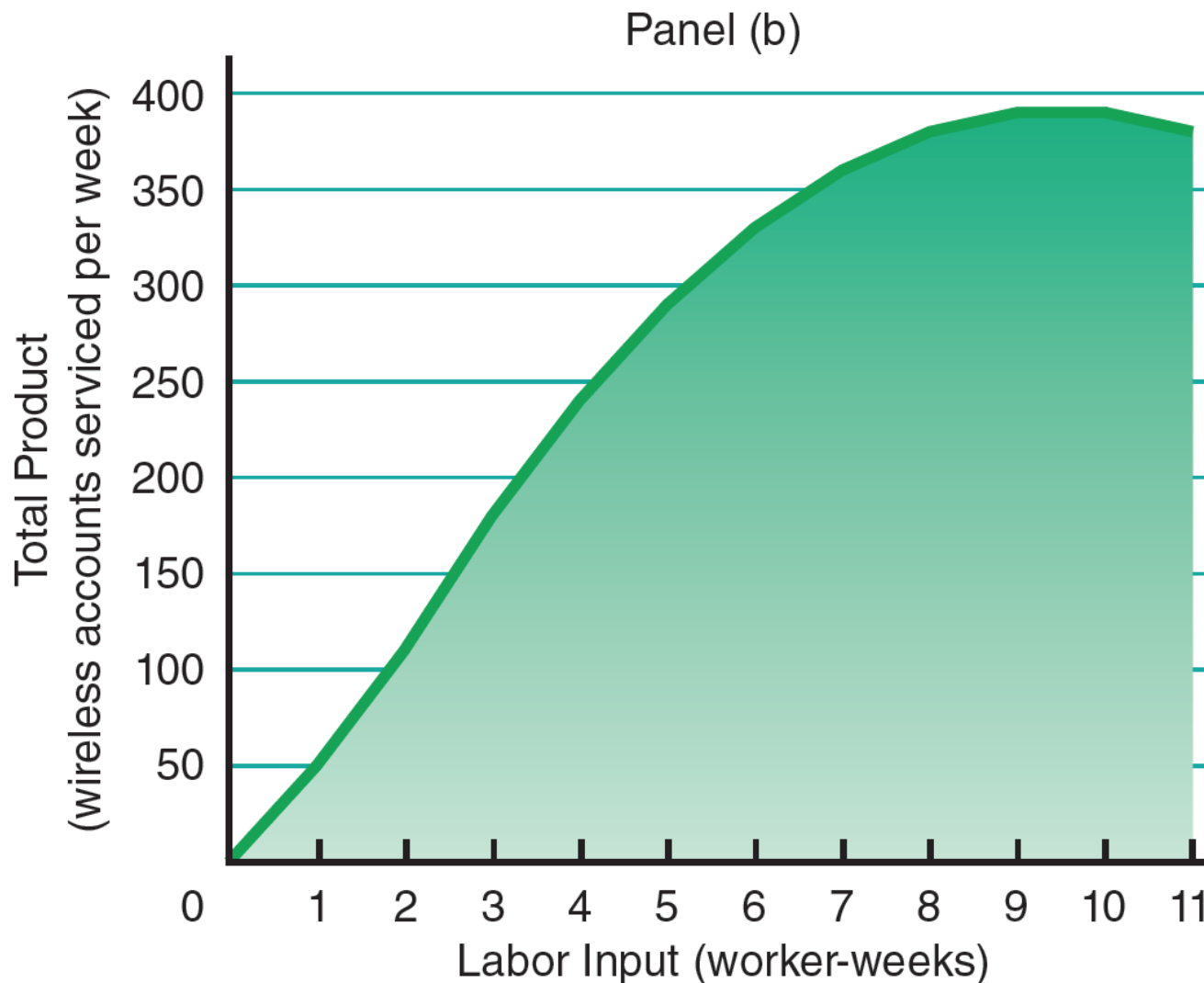
- The physical output that is due to the addition of one more unit of a variable factor of production
- The change in total product occurring when a variable input is increased and all other inputs are held constant
- Also called *marginal product*

The Production Relationship and Marginal Product: A Hypothetical Case, Panel (a)

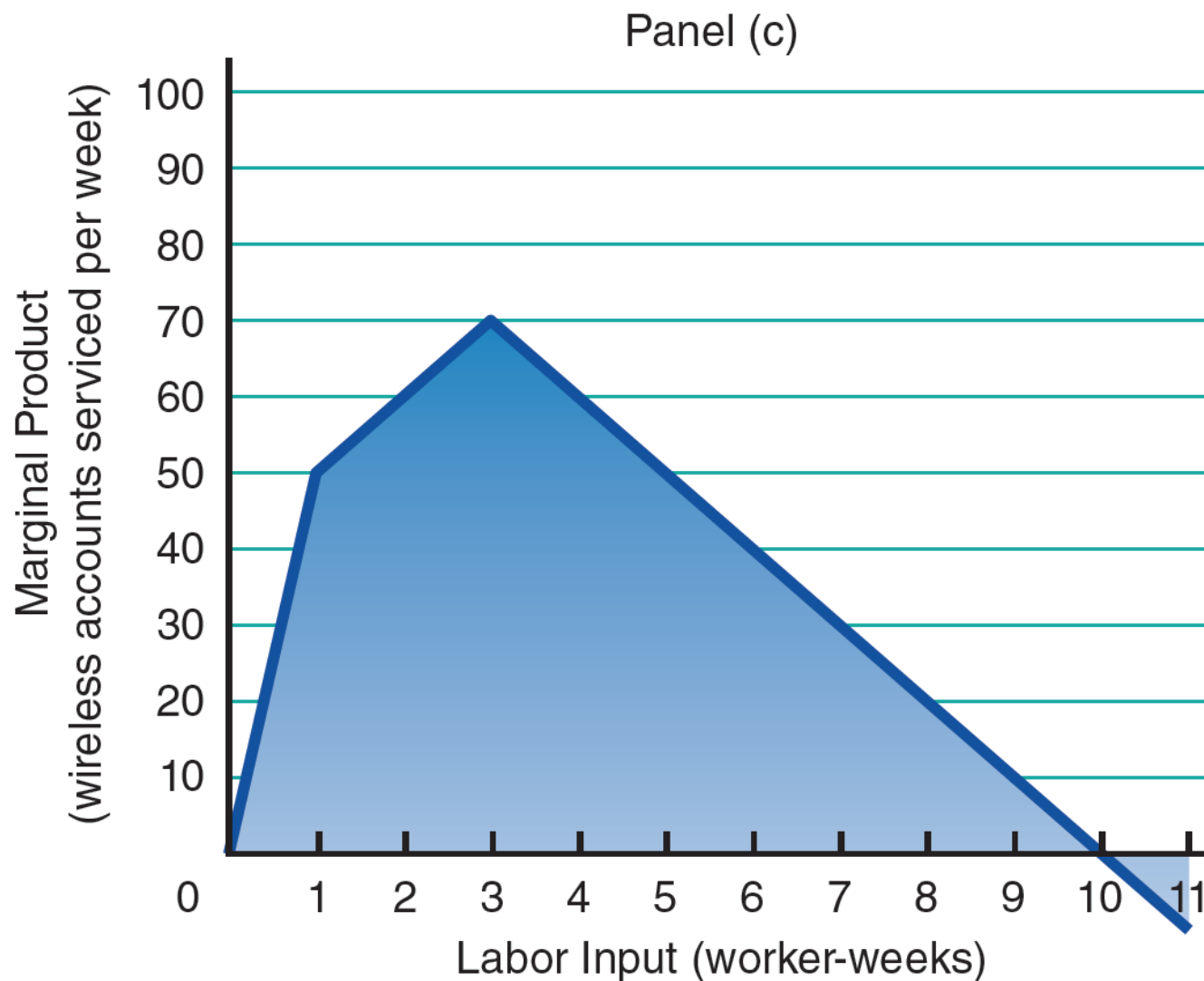
Panel (a)

(1)	(2)	(3)	(4)
Input of Labor (number of worker- weeks)	Total Product (output in units of wireless account services per week)	Average Product (total product ÷ number of worker-weeks) [units of wireless account services per week]	Marginal Product (output in units of wireless account services per week)
0	—	—	
1	50	50.00	50
2	110	55.00	60
3	180	60.00	70
4	240	60.00	60
5	290	58.00	50
6	330	55.00	40
7	360	51.43	30
8	380	47.50	20
9	390	43.33	10
10	390	39.00	0
11	380	34.55	-10

The Production Relationship and Marginal Product: A Hypothetical Case, Panel (b)



The Production Relationship and Marginal Product: A Hypothetical Case, Panel (c)



A Firm's Production (cont'd)

The marginal product of labor may increase very rapidly at the beginning.

Beyond some point, marginal product must begin to diminish as more workers are hired.

This is not because additional workers are less qualified but rather because each worker has, on average, fewer machines to work with.

A Firm's Production (cont'd)

Law of diminishing marginal product

- The observation that after some point, successive equal-sized increases in a variable factor of production, such as labor, added to fixed factors of production will result in smaller increases in output

A Firm's Production (cont'd)

Consider production of computer servers as an example of the law of diminishing marginal profit.

We have a fixed amount of factory space, assembly equipment, and quality control diagnostic software.

- The addition of more workers eventually yields successively smaller increases in output.

A Firm's Production (cont'd)

After a while, when all the assembly equipment and quality control diagnostic software are being used, additional workers will have to start assembling and troubleshooting quality problems manually.

The marginal physical product of an additional worker, given a specified amount of capital, must eventually be less than that for the previous workers.

A Firm's Production (cont'd)

Point of saturation:

- Given the amount of fixed inputs, there is no further positive use for more of the variable input.

Example: Food Service Firms Monitor Thrown-Out Food to Boost Marginal Product

Thrown-out food is a key contributor to diminishing marginal product for food service companies.

Some companies now use real-time monitoring devices to enable their managers to determine wastes in their food production processes.

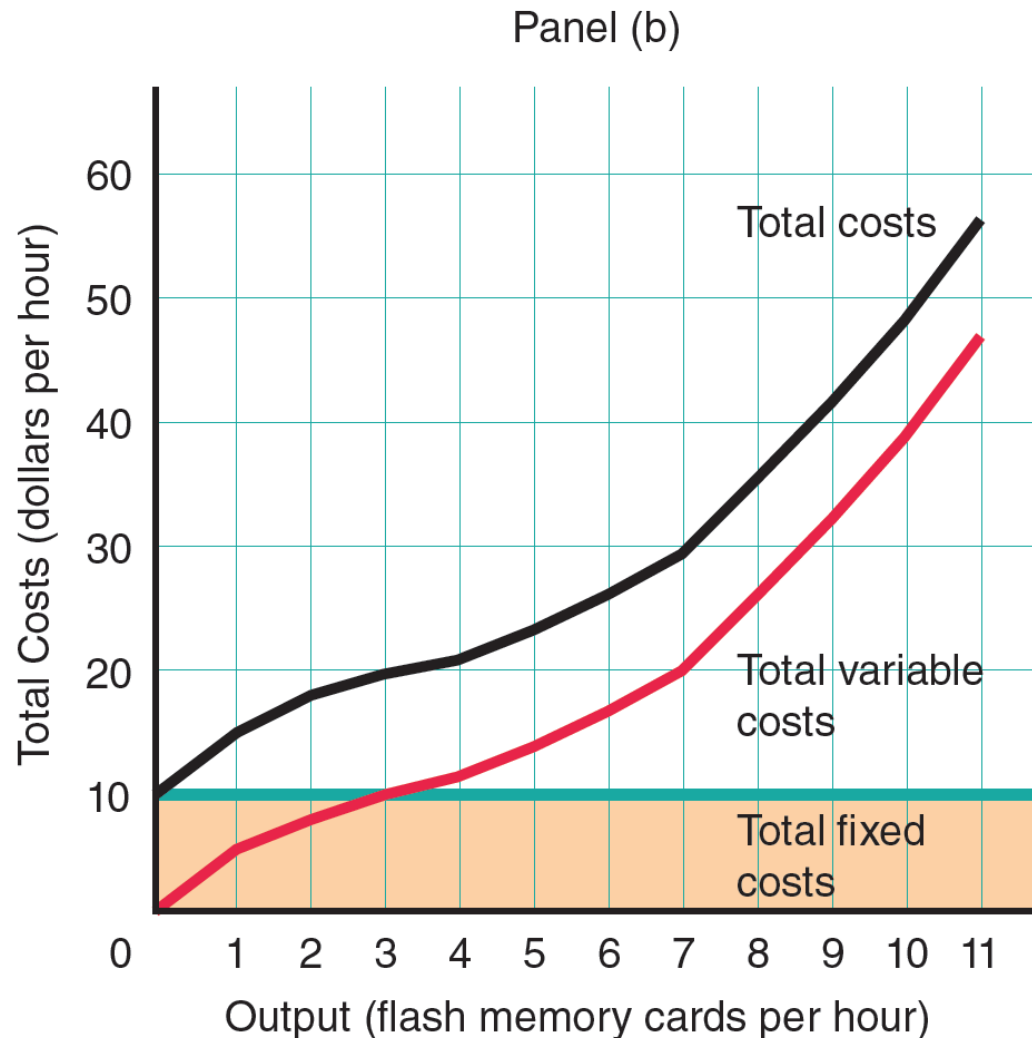
As a result, they can correct these processes in order to raise production per worker at any given level of food output.

Cost of Production: An Example, Panel (a)

Panel (a)

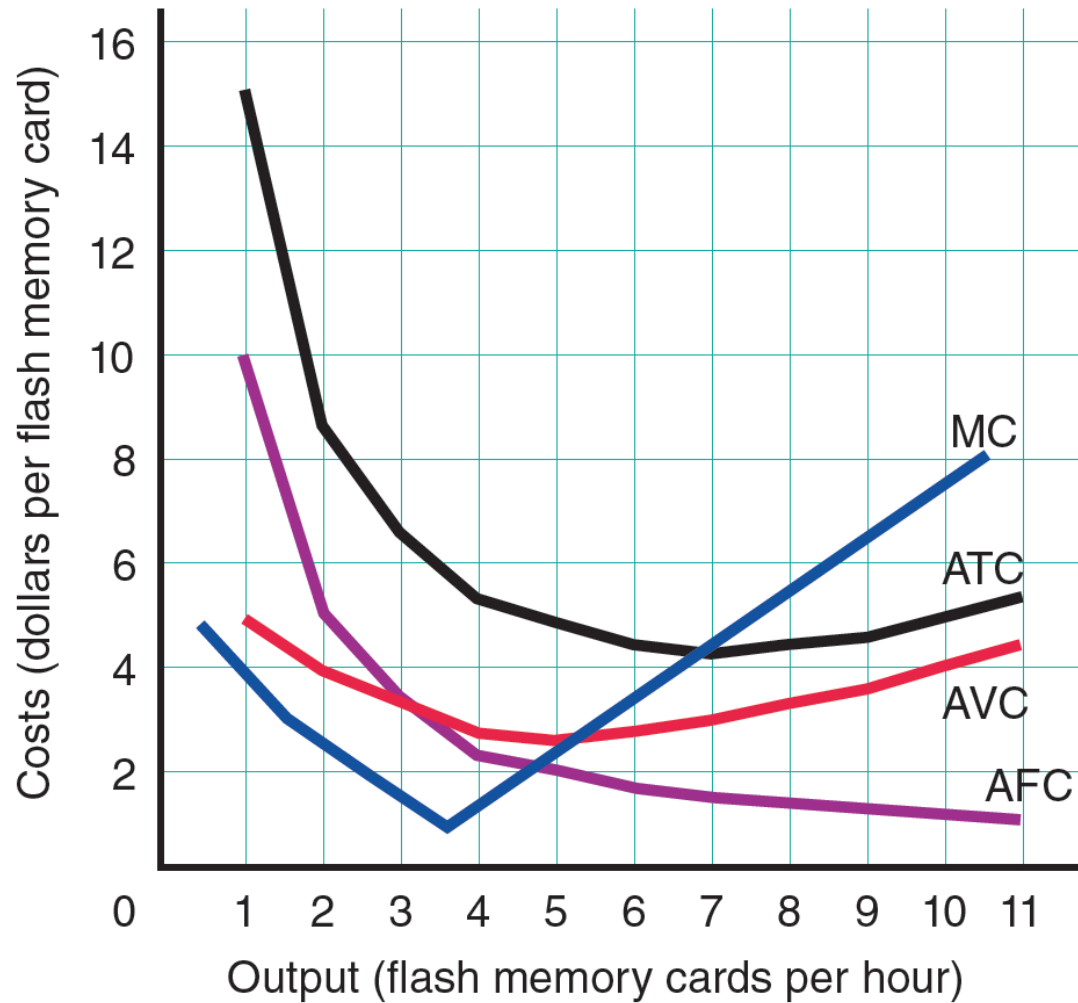
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Total Output (Q/hour)	Total Fixed Costs (TFC)	Total Variable Costs (TVC)	Total Costs (TC) (4) = (2) + (3)	Average Fixed Costs (AFC) (5) = (2) ÷ (1)	Average Variable Costs (AVC) (6) = (3) ÷ (1)	Average Total Costs (ATC) (7) = (4) ÷ (1)	Total Costs (TC) (4)	Marginal Cost (MC) (9) = $\frac{\text{Change in (8)}}{\text{Change in (1)}}$
0	\$10	\$ 0	\$10	—	—	—	\$10	
1	10	5	15	\$10.00	\$5.00	\$15.00	15	\$5
2	10	8	18	5.00	4.00	9.00	18	3
3	10	10	20	3.33	3.33	6.67	20	2
4	10	11	21	2.50	2.75	5.25	21	1
5	10	13	23	2.00	2.60	4.60	23	2
6	10	16	26	1.67	2.67	4.33	26	3
7	10	20	30	1.43	2.86	4.28	30	4
8	10	25	35	1.25	3.12	4.38	35	5
9	10	31	41	1.11	3.44	4.56	41	6
10	10	38	48	1.00	3.80	4.80	48	7
11	10	46	56	.91	4.18	5.09	56	8

Cost of Production: An Example, Panel (b)



Cost of Production: An Example, Panel (c)

Panel (c)



International Example: Global Airline Adoption of Tablet Devices to Save Jet Fuel and Cut Variable Costs

Most airlines have had their pilots replace paper manuals with tablet-based electronic manuals in order to save jet fuel.

Some airlines have been saving even more on fuel costs by replacing in-flight entertainment systems with tablet-based entertainment systems, which can cut a plane's weight by as much as 7 percent.

Short-Run Costs to the Firm

Total costs

- The sum of total fixed costs and total variable costs

Fixed costs

- Costs that do not vary with output and are fixed for a certain period of time, such as rent on a building

Variable costs

- Costs that vary with the rate of production, such as wages paid to workers and purchases of materials

$$\text{Total costs (TC)} = \text{TFC} + \text{TVC}$$

Short-Run Costs to the Firm (cont'd)

Average total costs (ATC)

- Also called average per-unit total costs

$$\text{Average total costs (ATC)} = \frac{\text{Total costs (TC)}}{\text{Output (Q)}}$$

Short-Run Costs to the Firm (cont'd)

Average variable costs (AVC)

$$\text{Average variable costs (AVC)} = \frac{\text{Total variable costs (TVC)}}{\text{Output (Q)}}$$

Short-Run Costs to the Firm (cont'd)

Average fixed costs (AFC)

$$\text{Average fixed costs (AFC)} = \frac{\text{Total fixed costs (TFC)}}{\text{Output (Q)}}$$

International Example: New Pallet Materials Reduce Firms' Average Total Costs

Increasingly, shipping companies have been replacing wood pallets with lighter materials, such as plastics, corrugated cardboard, and metal.

Because of their smaller volumes, these new materials also allow shippers to stack more loads within the same space on trucks, ships, and planes.

As a result, the average total costs incurred in shipping are reduced.

Short-Run Costs to the Firm (cont'd)

Marginal cost

- The change in total costs due to a one-unit change in production rate

$$\text{Marginal cost (MC)} = \frac{\text{Change in total cost}}{\text{Change in output}}$$

Short-Run Costs to the Firm (cont'd)

Question:

- Do you think there is a predictable relationship between the production function and AVC, ATC, and MC?

Short-Run Costs to the Firm (cont'd)

Answer:

- As long as marginal physical product rises, marginal cost will fall, and when marginal physical product starts to fall (after reaching the point of diminishing marginal product), marginal cost will begin to rise.

Short-Run Costs to the Firm (cont'd)

The relationship between average and marginal costs:

- When marginal costs are less than average variable costs, the latter must fall.
- When marginal costs are greater than average variable costs, the latter must rise.

Short-Run Costs to the Firm (cont'd)

There is also a relationship between marginal costs and average total costs:

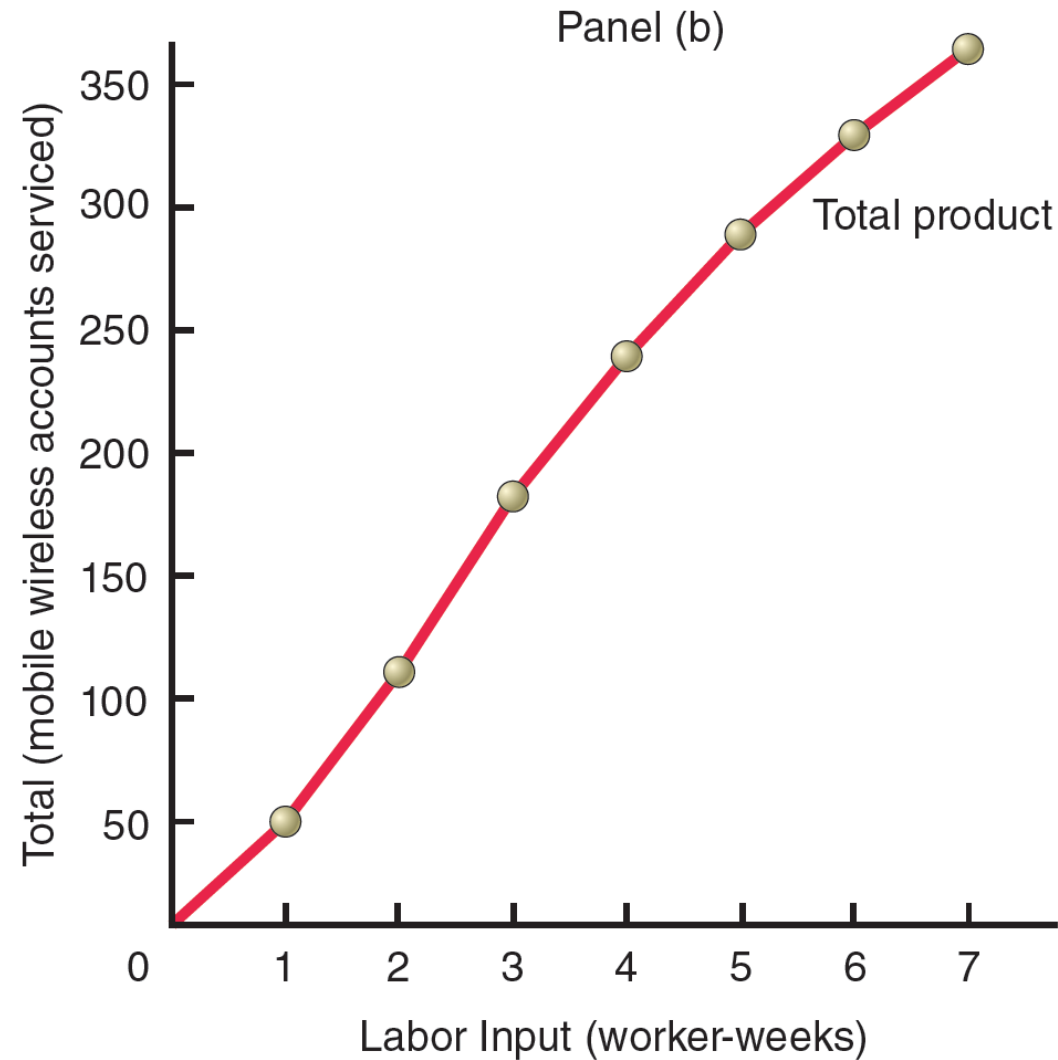
- Average total cost is equal to total cost divided by the number of units produced.
- Marginal cost is the change in total cost due to a one-unit change in the production rate.

Figure 22-3 The Relationship between Output and Costs, Panel (a)

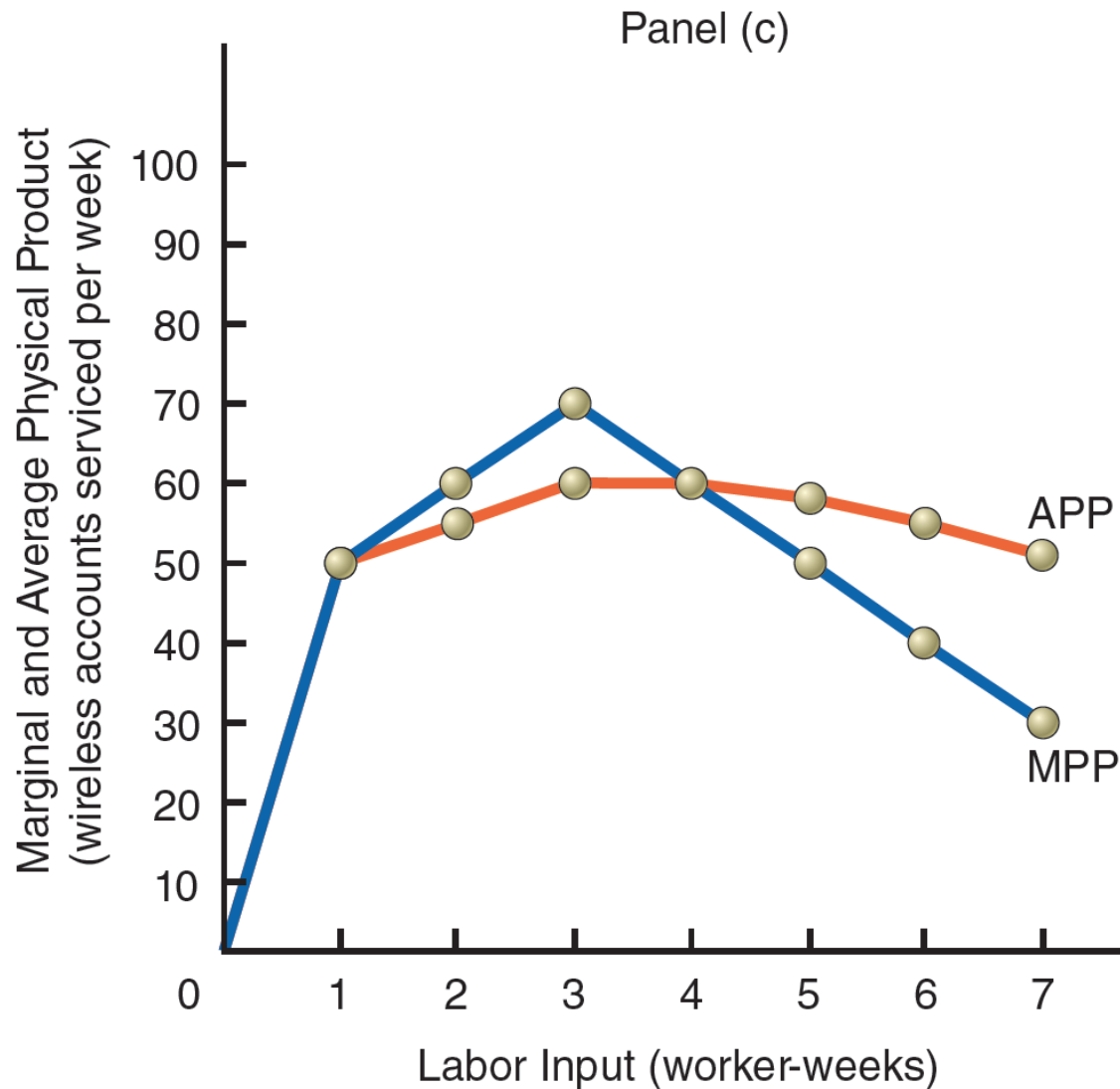
Panel (a)

(1) Labor Input	(2) Total Product (number of wireless accounts serviced per week)	(3) Average Physical Product (accounts per week) $(3) = (2) \div (1)$	(4) Marginal Physical Product	(5) Average Variable Cost $(5) =$ $\$1,000 \div (3)$	(6) Marginal Cost $(6) =$ $\$1,000 \div (4)$
0	0	—	—	—	—
1	50	50	50	\$20.00	\$20.00
2	110	55	60	18.18	16.67
3	180	60	70	16.67	14.29
4	240	60	60	16.67	16.67
5	290	58	50	17.24	20.00
6	330	55	40	18.18	25.00
7	360	51	30	19.61	33.33

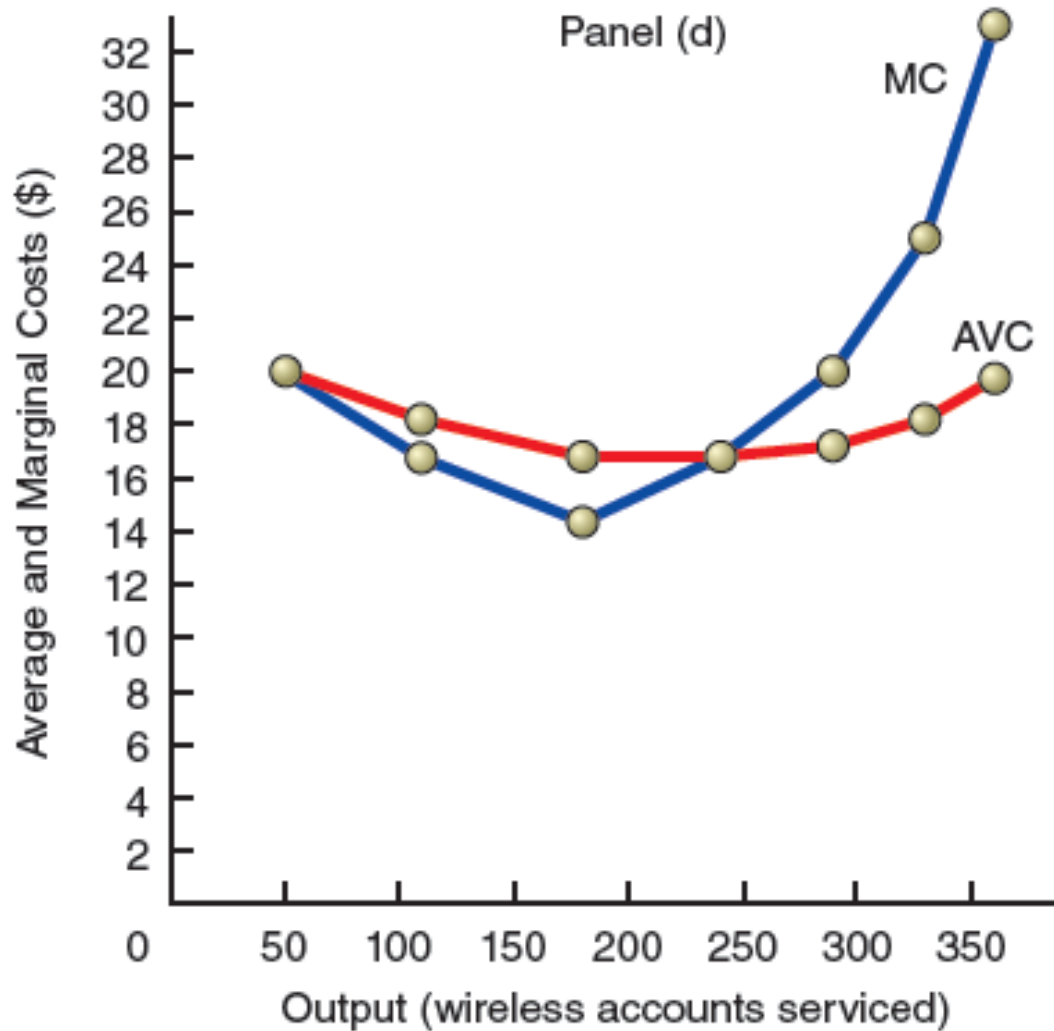
The Relationship between Output and Costs, Panel (b)



The Relationship between Output and Costs, Panel (c)



The Relationship between Output and Costs, Panel (d)



Short-Run Costs to the Firm (cont'd)

Firms' short-run cost curves are a reflection of the law of diminishing marginal product.

Given any constant price of the variable input, marginal costs decline as long as the marginal product of the variable resource is rising.

22.3 Short-Run Costs to the Firm (cont'd)

At the point at which diminishing marginal product begins, marginal costs begin to rise as the marginal product of the variable input begins to decline.

Short-Run Costs to the Firm (cont'd)

If the wage rate is constant, then the labor cost associated with each additional unit of output will decline as long as the marginal physical product (MPP) of labor increases.

Short-Run Costs to the Firm (cont'd)

Of course, the average total cost curve and average variable cost curve are also affected.

They will have their familiar U shape in the short run.

Long-Run Cost Curves

Planning horizon

- The long run, during which all inputs are variable

Long-Run Cost Curves (cont'd)

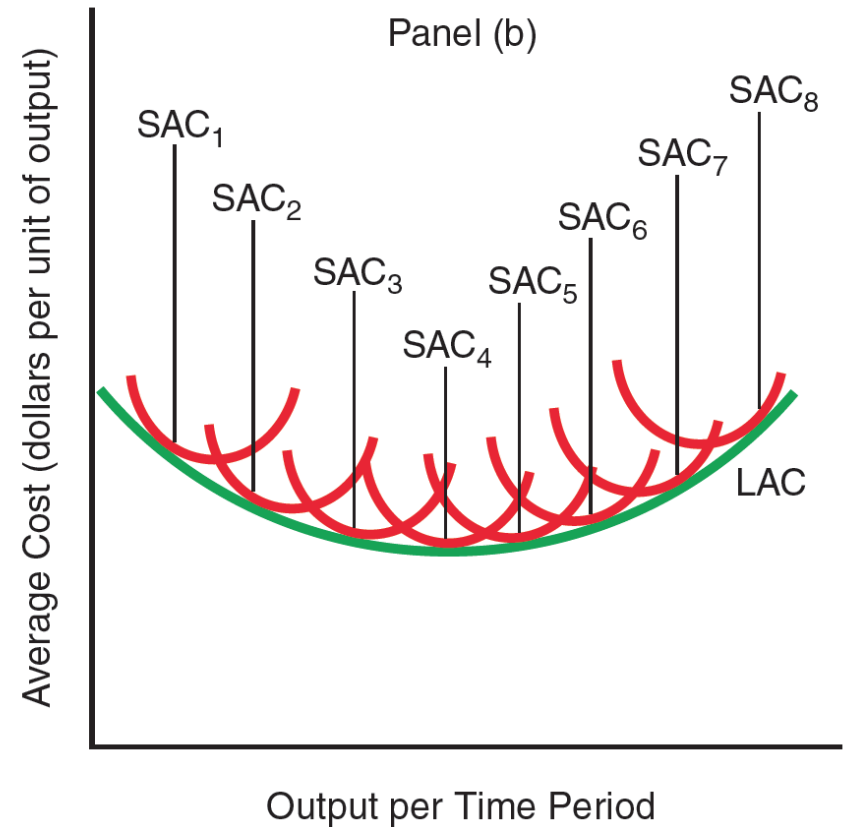
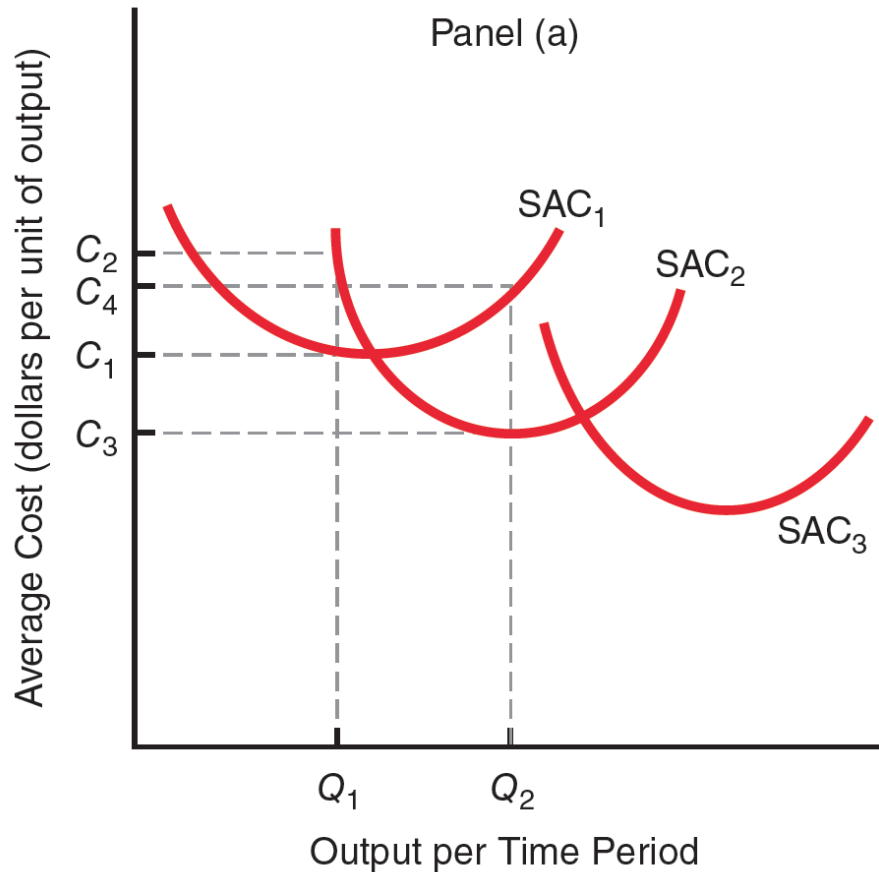
Long-run average cost curve

- The locus of points representing the minimum unit cost of producing any given rate of output, given current technology and resource prices

Planning curve

- The long-run average cost curve

Preferable Plant Size and the Long-Run Average Cost Curve



Long-Run Cost Curves (cont'd)

Observation:

- Only at the minimum point of the long-run average cost curve is short-run average cost curve tangent to long-run average cost curve.

Question:

- Why do you think the long-run average cost curve is U-shaped?

Long-Run Cost Curves (cont'd)

Economies of scale

Constant returns to scale

Diseconomies of scale

Long-Run Cost Curves (cont'd)

Economies of scale

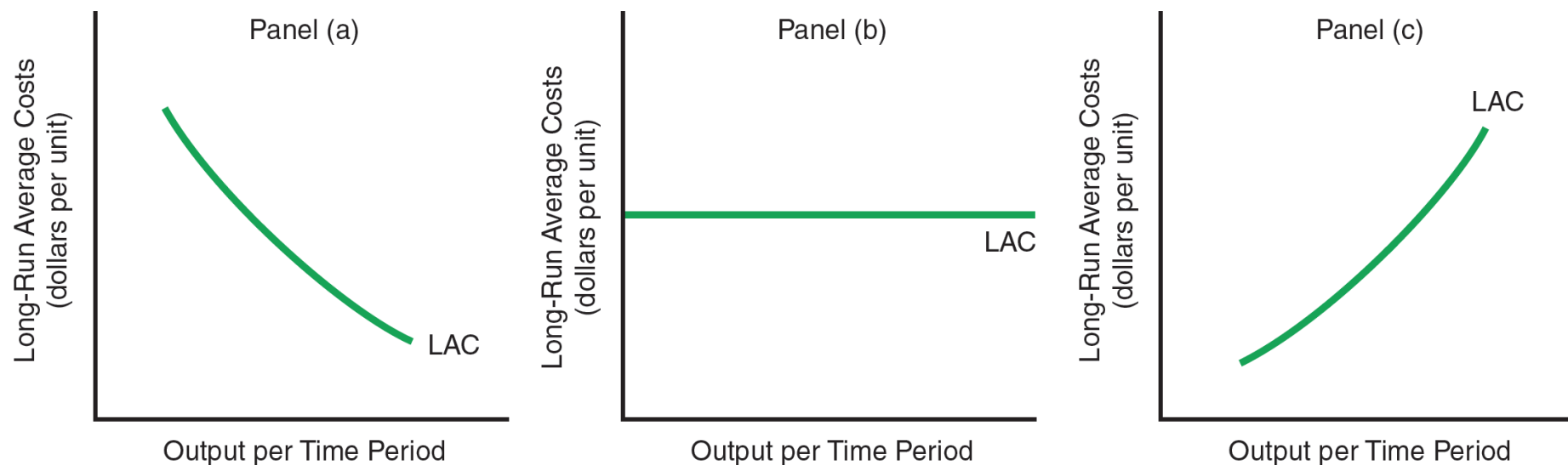
- Decreases in long-run average costs resulting from increases in output
 - These economies of scale do not persist indefinitely, however.
 - Once long-run average costs rise, the curve begins to slope upward.

Long-Run Cost Curves (cont'd)

Reasons for economies of scale:

- Specialization:
 - Division of tasks or operations
- Dimensional factor:
 - Large-scale firms often require proportionately less input per unit of output
- Improved productive equipment:
 - The larger the enterprise, the more the firm can take advantage of larger-volume types of machinery.

Economies of Scale, Constant Returns to Scale, and Diseconomies of Scale Shown with the Long-Run Average Cost Curve



Long-Run Cost Curves (cont'd)

Explaining why a firm might experience diseconomies of scale.

- There are limits to the efficient functioning of management.
- Coordination and communication is more challenging as firm size increases.

International Example: Maersk Super-Scale Ship Threatens to Sink under Diseconomies

The Danish shipping firm Maersk has recently built 20 of the largest ships ever made, the Triple-E class, which can carry about 11 percent more containers than the second-largest freight vessels.

However, the Triple-E's bulk cannot fit into any port in the Western Hemisphere, and it is too wide to pass through the recently widened Panama Canal.

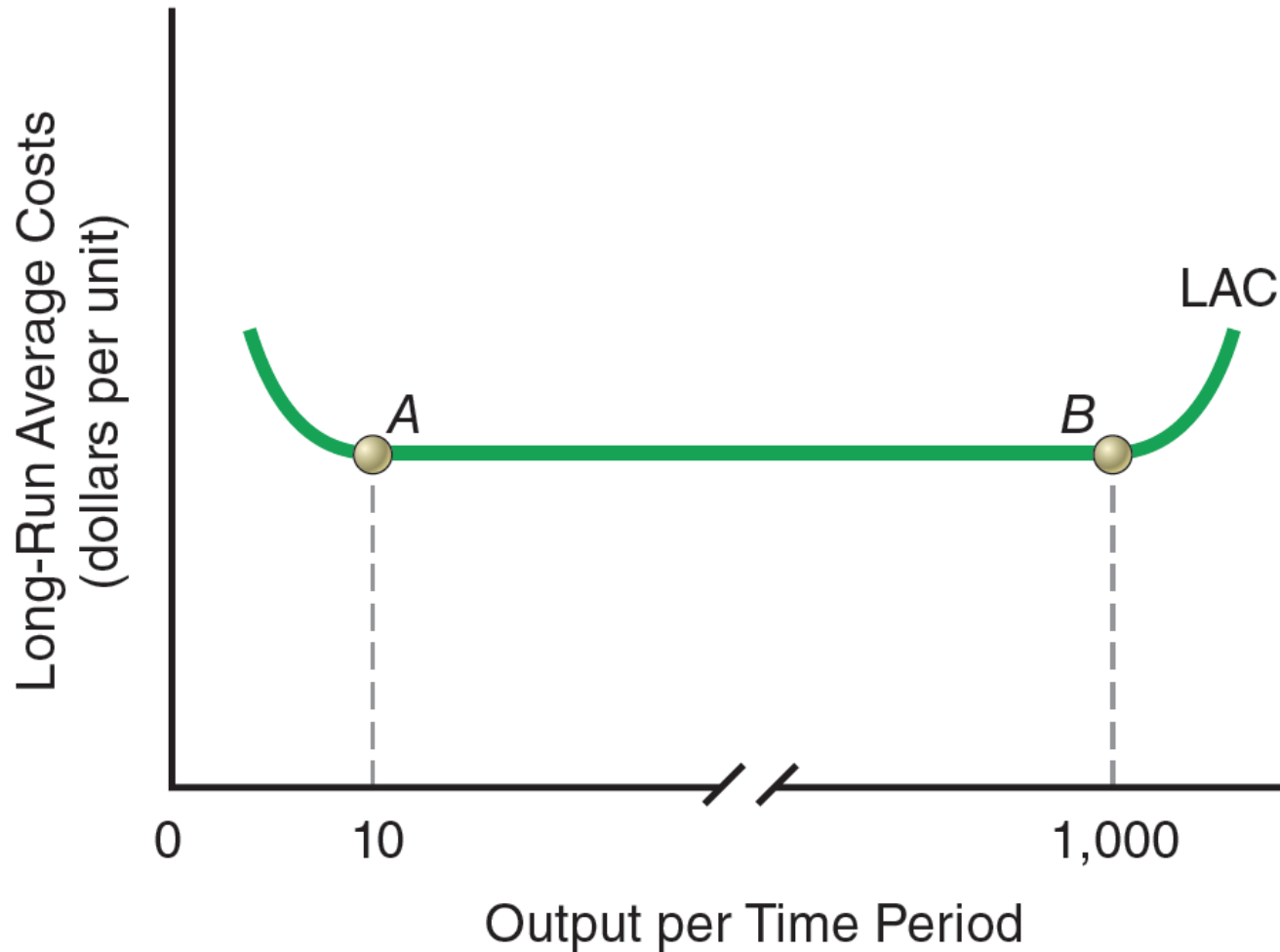
So the average cost of operating a Triple-E is higher than for Maersk's other ships, implying its *diseconomies of scale*.

Long-Run Cost Curves (cont'd)

Minimum efficient scale (MES)

- The lowest rate of output per unit time at which long-run average costs for a particular firm are at a minimum

Minimum Efficient Scale



Long-Run Cost Curves (cont'd)

Small MES relative to industry demand:

- Room for many efficient firms
- High degree of competition

Large MES relative to industry demand:

- Room for only a small number of efficient firms
- Small degree of competition

What If ... the government decided that most U.S. companies were “too large” and legally required them to downsize their operations?

Forcing firms to downsize over their planning horizons would force them to produce a long-run output rate less than the MES.

Each firm's average cost would exceed the minimum feasible average cost.

Reducing the Fixed Costs of Space Exploration

Bigelow Aerospace will replace the traditional spacecraft for space exploration with an “inflatable space pod” made of light and soft materials.

The cost of launching this new pod will reduce, and once in orbit, the pod can be used by different crews over time.

So the new pod promises to lower the fixed costs of orbital space exploration.

END OF CHAPTER